Exercises 8.B

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Exercise 1. Suppose V is a complex vector space, $N \in \mathcal{L}(V)$, and 0 is the only eigenvalue of N. Prove that N is nilpotent.

Proof. By 8.21,

$$\begin{aligned} V &= G(0, N) \\ &= \operatorname{null}(N - 0I)^{\dim V} \\ &= \operatorname{null} N^{\dim V}, \end{aligned}$$

so given $v \in V$, we have $N^{\dim V} v = 0$, thus N is nilpotent. $\square$

Exercise 2. Give an example of an operator T on a finite-dimensional real vector space such that 0 is the only eigenvalue of T but T is not nilpotent.

Proof. Let $T \in \mathbb{R}^3$ be defined by $T(z_1, z_2, z_3) = (-z_2, z_1, 0)$. Then the only eigenvalue of T is 0, but $T^3(z_1, z_2, z_3) = (z_2, -z_1, 0)$, so T is not nilpotent. $\square$

Exercise 3. Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible. Prove that T and $S^{-1}TS$ have the same eigenvalues with the same multiplicities.

Proof. We seek to show that $S|_{G(\lambda, S^{-1}TS)}$ is in fact an isomorphism from $G(\lambda, S^{-1}TS)$ to $G(\lambda, T)$ for $\lambda \in \mathbf{F}$, which shows that when the former is nontrivial (so that λ is an eigenvalue of $S^{-1}TS$), so is the latter, and vice versa. We already know that $S|_{G(\lambda, S^{-1}TS)}$ is invertible since S is invertible, so it is an isomorphism onto its range. Thus we need only show that $G(\lambda, T) = \operatorname{range} S|_{G(\lambda, S^{-1}TS)}$.

We know that $(S^{-1}TS - \lambda I)^{\dim V} = S^{-1}(T - \lambda I)^{\dim V} S$ since $p(S^{-1}TS) = S^{-1}p(T)S$ for any p, T, and invertible S. For a given eigenvalue λ of $S^{-1}TS$, let $v \in G(\lambda, S^{-1}TS)$. Then

$$\begin{aligned} 0 &= (S^{-1}TS - \lambda I)^{\dim V} v \\ &= S^{-1}(T - \lambda I)^{\dim V} Sv \\ &= (T - \lambda I)^{\dim V} Sv, \end{aligned}$$

where the final equality comes from applying S to both sides on the left. Thus $Sv \in G(\lambda, T)$.

Likewise, given $u \in G(\lambda, T)$, we can express u uniquely as $u = Sv$ for some $v \in V$ since S is surjective, so

$$\begin{aligned} 0 &= (T - \lambda I)^{\dim V} u \\ &= (T - \lambda I)^{\dim V} Sv \\ &= S^{-1}(T - \lambda I)^{\dim V} Sv \\ &= (S^{-1}TS - \lambda I)^{\dim V} v, \end{aligned}$$

where the third equality comes from applying S^{-1} to both sides on the left. Thus $v \in G(\lambda, S^{-1}TS)$.

This means S is an isomorphism from $G(\lambda, S^{-1}TS)$ to $G(\lambda, T)$, so

$$\dim G(\lambda, S^{-1}TS) = \dim G(\lambda, T).$$

□

Exercise 4. Suppose V is an n-dimensional complex vector space and T is an operator on V such that $\operatorname{null} T^{n-2} \neq \operatorname{null} T^{n-1}$. Prove that T has at most two distinct eigenvalues.

Proof. □